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ECEN 5224

Week 4 Lab Report

02/18/26

Lab 1

Objective:

The objective of this lab was to use ADS to run TDR simulations and to understand the characteristic impedance of a microstrip and a stripline transmission line. The results are graphed and compared to other PCB impedance calculators.

Method:

ADS was used to simulate a microstrip and stripline with properties of: h = 60 mils, w = 50 mils, t = 1.4 mils and Dk = 4.8 for the microstrip and h = 10 mils, w = 10 mils, t = 0.7 mils and Dk = 4.6 for the stripline. Time-domain TDR simulations were run to plot the impedances and were then compared with results from Saturn PCB and Sierra Circuits impedance calculators.

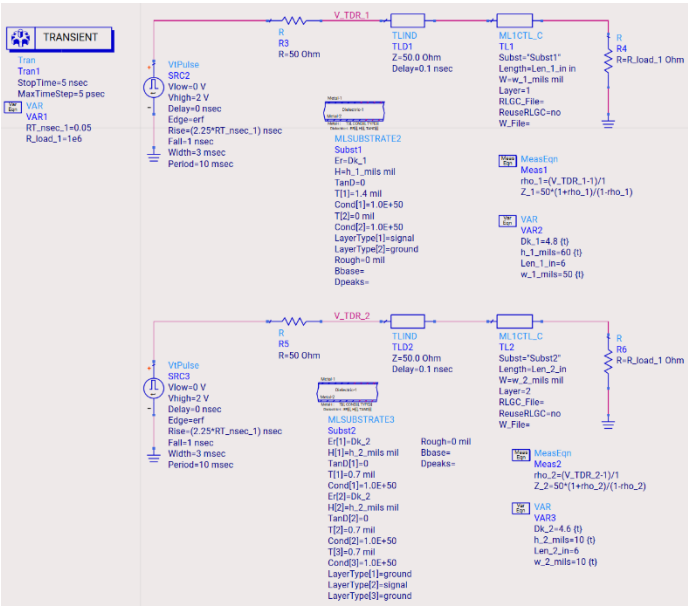


Figure 1: Lab 1 Completed Circuit

Results and Plots:

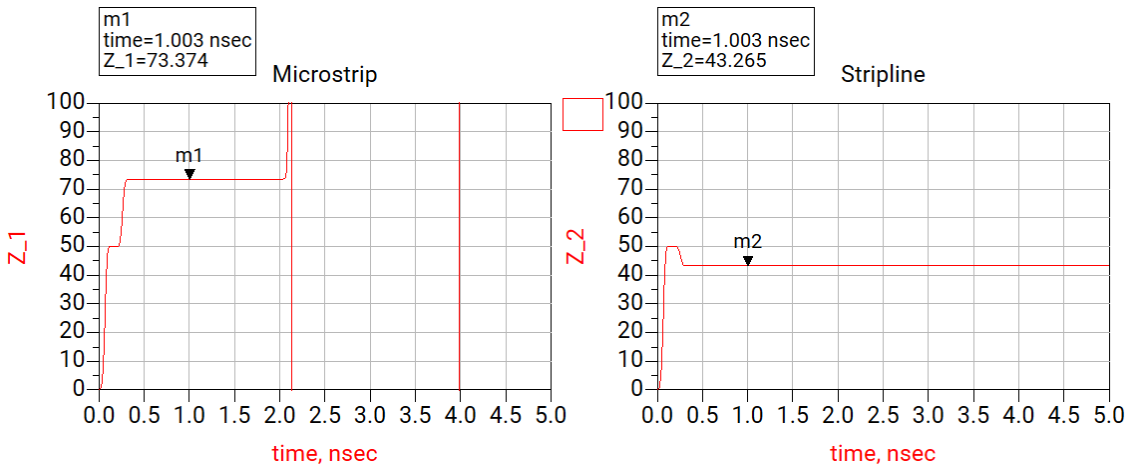


Figure 2: Impedance of Microstrip and Stripline

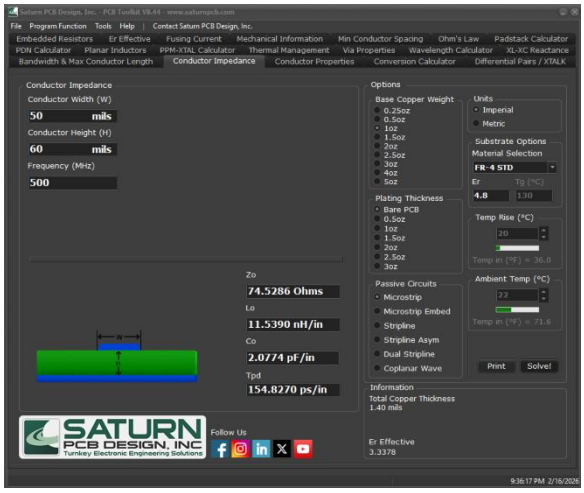


Figure 3: Saturn PCB Microstrip

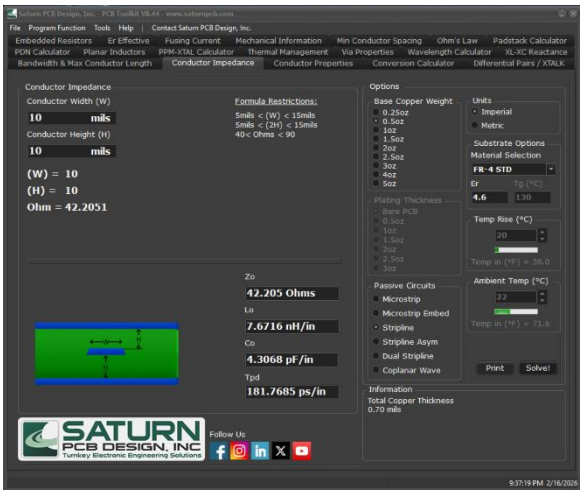


Figure 4: Saturn PCB Stripline

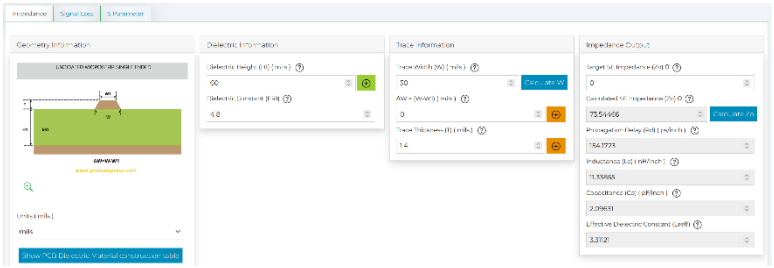


Figure 5: Sierra Circuits Microstrip

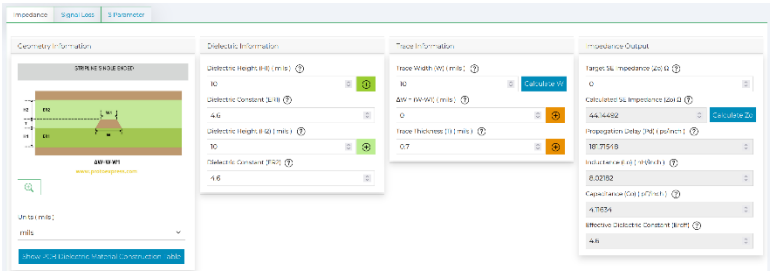


Figure 6: Sierra Circuits Stripline

Analysis and Conclusion:

From the ADS simulations, the microstrip showed an impedance of approximately 73.374 Ω, while the stripline show an impedance of approximately 43.265 Ω. These values match with the results seen from the Sierra Circuits and Saturn PCB calculators, which predicted approximately 73.5 - 74.5 Ω for the microstrip and around 42.2 - 44.1 Ω for the stripline. Although not exactly the same as the ADS simulation, all impedances match up well within an acceptable range.\

The ADS TDR simulations successfully graph the impedances of the microstrip and stripline transmission lines. The captured impedance measurements strongly compare with the various impedance calculators. This supports the data of TDR-based impedance and shows the importance of consistent parameters when recording data in an analytical or simulation-based way.

Lab 2

Objective:

The objective of this lab is to use ADS simulate the signal at the RX in a point-to-point transmission line circuit with various different termination methods.

Method:

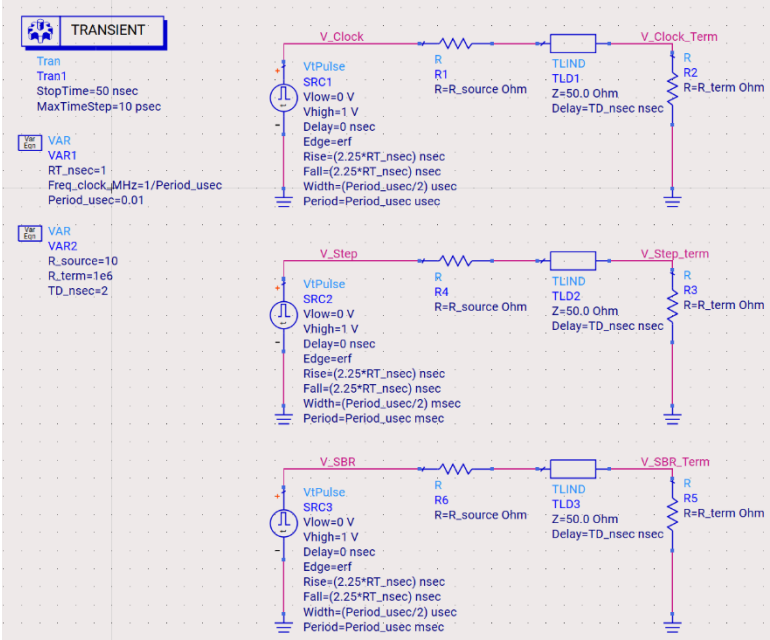


Figure 7: Lab 2 Completed Circuit

A 50 Ω transmission line with a 10 Ω driver impedance, 1 ns rise time, and 2 ns delay was run with a transient simulation performed using clock, step, and single-bit response voltage sources to observe the received waveform under four terminations. The four cases: no termination, source series termination, far-end to ground termination, and far-end to Vtt termination. The vtt termination method uses the same circuit as above with a DC voltage source of 0.5 V beneath the R_term resistor.

Results and Plots:

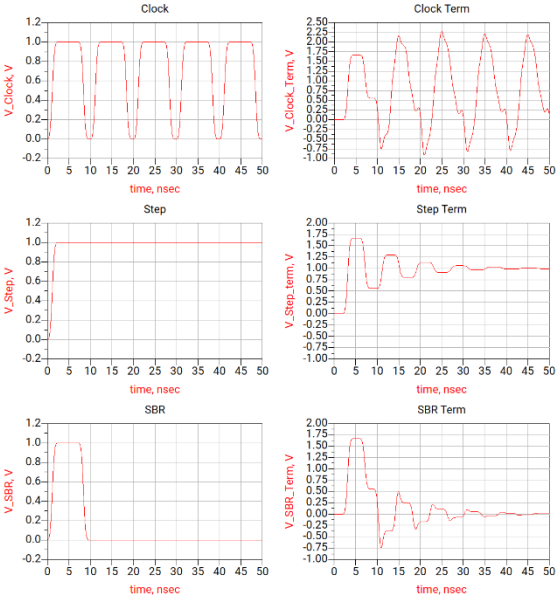


Figure 8: No Termination

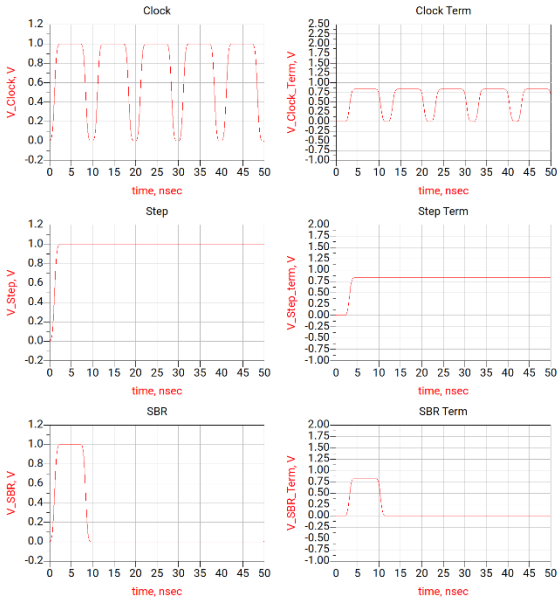


Figure 9: Far End to GND Termination

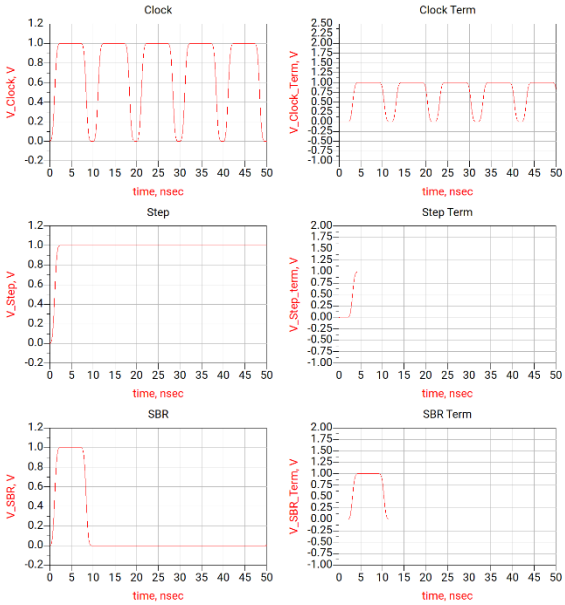


Figure 10: Source Series Termination

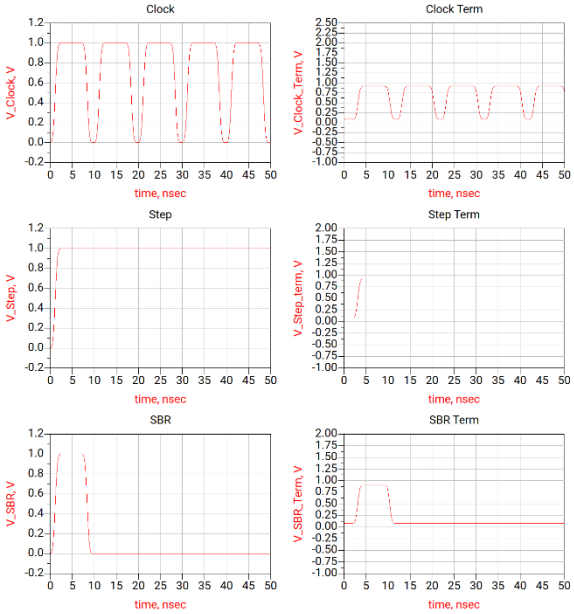


Figure 11: Far End to Vtt Termination

Analysis and Conclusion:

Transient simulations of a 50 Ω transmission line with a 10 Ω driver impedance, 1 ns rise time, and 2 ns delay showed that different terminations change the RX waveform for clock, step, and SBR sources. With no termination, the measured wave illustrated ringing from reflections due to the high-impedance load. This is seen best in the clock and SBR graphs. With source series termination the wave's response became much more clean. The reflections returning to the source were absorbed instead of reflected. This lead to far less ringing and a more stable waveform. With far-end to ground termination, the wave shows even less reflections, but the steady-state level decreased to approximately around 0.83 V. This is due to the voltage divider between the 10 Ω driver and the matched 50 Ω transmission line. The far-end to Vtt termination, with Vtt being 0.5 V, produced a waveform similar to the results of the far-end to ground termination. The steady-state level appeared to have risen slightly to ~0.9 V.

These simulations demonstrated that impedance matching can impact reflection behavior and leading to better or worse signal quality in point-to-point transmission lines. Lines without a termination produce significant ringing while source and far-end terminations have far more stable waves. The far-end matching provides the best reflection damping with Vtt termination showing similar performance but with a higher steady-state.